Applied Mathematics for Chemists

(화학수학)

Lecture 15

Operators

Mathematical Operators (연산자)

perform mathematical operations on some function

e.g. "+3" operator, "×2" operator, "$\mathrm{d}/\mathrm{d}x$" operator

- Typically denoted by a letter with a caret (^)
- Operate on the function on its right

Special Operators

- Identity operator (항등 연산자) $\hat{E}$: multiplication by 1 (doing nothing)

$$\hat{E}f = f$$

- Null operator (영 연산자) $\hat{0}$: multiplication by 0

$$\hat{0}f = 0$$

Operator Algebra

- Sum of two operators: if $\hat{C} = \hat{A} + \hat{B}$, then

$$\hat{C}f = \left(\hat{A} + \hat{B}\right)f = \hat{A}f + \hat{B}f$$

- Product of two operators: if $\hat{C} = \hat{A}\hat{B}$, then

$$\hat{C}f = \hat{A}\left(\hat{B}f\right)$$

Properties of Operator Multiplication

- Associative property (결합법칙)

$$\left(\hat{A}\hat{B}\right)\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property (분배법칙)

$$\hat{A}\left(\hat{B} + \hat{C}\right) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property (교환법칙) → Not necessarily

$$\hat{A}\hat{B} = \hat{B}\hat{A}\ ?$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to commute

- Commutator (교환자)

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

Exponentiation of Operator

(연산자의 거듭제곱)

$$\hat{A}^n = \hat{A}\hat{A}\cdots\hat{A} \text{ (} n \text{ times)}$$

Inverse of Operator (역 연산자)

- $\hat{A}^{-1}$: inverse of operator $\hat{A}$

$$\hat{A}\hat{A}^{-1} = \hat{A}^{-1}\hat{A} = \hat{E}$$

- $\hat{A}$ is also the inverse of $\hat{A}^{-1}$

- Not all operators have inverses (*e.g.* $\hat{0}$)

Operator Algebra and Differential Equation

- Derivative operator $\hat{\mathrm{D}}_x = \mathrm{d}/\mathrm{d}x$

- Homogeneous linear DE with constant coefficients

$$a_n \frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1} \frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1 \frac{\mathrm{d}y}{\mathrm{d}x} + a_0 y = 0$$

$$\left(a_n \hat{\mathrm{D}}_x^n + a_{n-1} \hat{\mathrm{D}}_x^{n-1} + \cdots + a_1 \hat{\mathrm{D}}_x + a_0 \hat{E}\right) y = 0$$

This can be solved by operator algebra

Eigenfunctions and Eigenvalues
(고유함수와 고윳값)

For a given operator $\hat{A}$, a function f such that

$$\hat{A}f = af$$

is called an eigenfunction of $\hat{A}$, and a is called an eigenvalue. This equation is called an eigenvalue equation (고윳값 방정식).

Operators in Quantum Mechanics

"For every observable quantity,

there is a corresponding mathematical operator"

- Position operators: $\hat{x} = x,\ \hat{y} = y,\ \hat{z} = z$
- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x}, \qquad \hat{p}_y = \frac{\hbar}{i}\frac{\partial}{\partial y}, \qquad \hat{p}_z = \frac{\hbar}{i}\frac{\partial}{\partial z}$$

- System energy operator (Hamiltonian):

$$\widehat{H} = -\frac{\hbar}{2m}\nabla^2 + \mathcal{V}(x, y, z)$$

Hermitian Property (에르미트 성질)

- If $\hat{A}$ is Hermitian, it obeys the relation

$$\int \chi^*(\hat{A}\psi)\,\mathrm{d}q = \int (\hat{A}\chi)^*\psi\,\mathrm{d}q$$

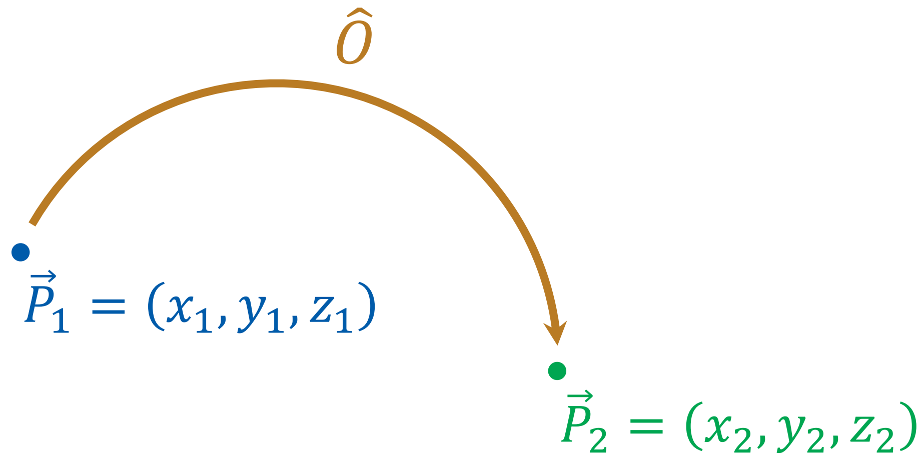
where χ and ψ are arbitrary functions and q represents all of the coordinates.

- All of the quantum-mechanical operators are Hermitian

Today’s Class

- Symmetry operators
  - Point symmetry operators
  - Symmetry elements
  - Symmetry of objects
  - Symmetry operators on functions
  - Eigenfunctions of symmetry operators

Symmetry Operators



$$\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\hat{O}\vec{P}_1 = \vec{P}_2$$

Point Symmetry Operators
(점대칭 연산자)

a class of symmetry operators
that leave at least one point unmoved
(the unmoved point = origin)

Identity operator

Inversion operator

Reflection operator

Rotation operator

Identity Operator and Inversion Operator

(항등 연산자 및 반전 연산자)

- Identity operator $\hat{E}$: doing nothing

$$\hat{E}(x_1, y_1, z_1) = (x_1, y_1, z_1)$$

$$\hat{E}\vec{r}_1 = \vec{r}_1$$

- Inversion operator $\hat{\imath}$: inversion w.r.t. the origin

$$\hat{\imath}(x_1, y_1, z_1) = (-x_1, -y_1, -z_1)$$

$$\hat{\imath}\vec{r}_1 = -\vec{r}_1$$

Reflection Operators (반사 연산자) $\hat{\boldsymbol{\sigma}}$

reflection through a plane

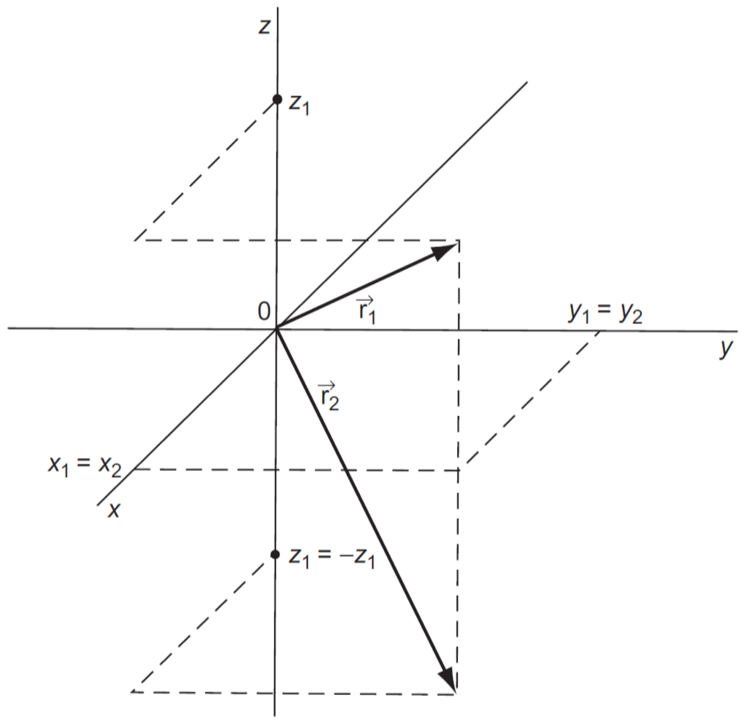
(The plane must be specified)

$$\hat{\sigma}_{(xy)}(x_1, y_1, z_1) = (x_1, y_1, -z_1)$$

$$\hat{\sigma}_{(xz)}(x_1, y_1, z_1) = (x_1, -y_1, z_1)$$

$$\hat{\sigma}_{(yz)}(x_1, y_1, z_1) = (-x_1, y_1, z_1)$$

Action of $\hat{\boldsymbol{\sigma}}_{(xy)}$



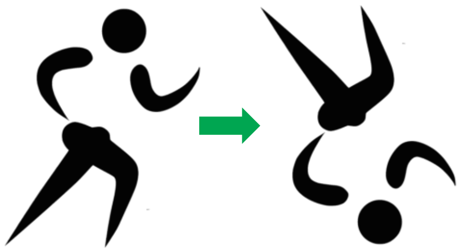
(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.1)

Horizontal and Vertical Reflections

If the system has a principal axis,

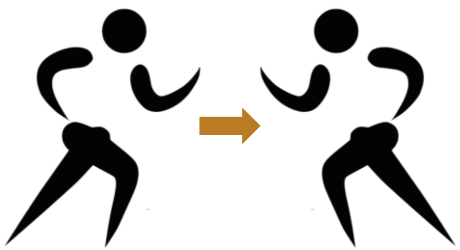
we normally regard it as the z axis

- Horizontal reflection $\hat{\sigma}_{\mathrm{h}} = \hat{\sigma}_{(xy)}$ (unique)



- Vertical reflection $\hat{\sigma}_{\mathrm{v}}$: other reflections

$$\hat{\sigma}_{\mathrm{v}(xz)}, \hat{\sigma}_{\mathrm{v}(yz)}, \cdots$$



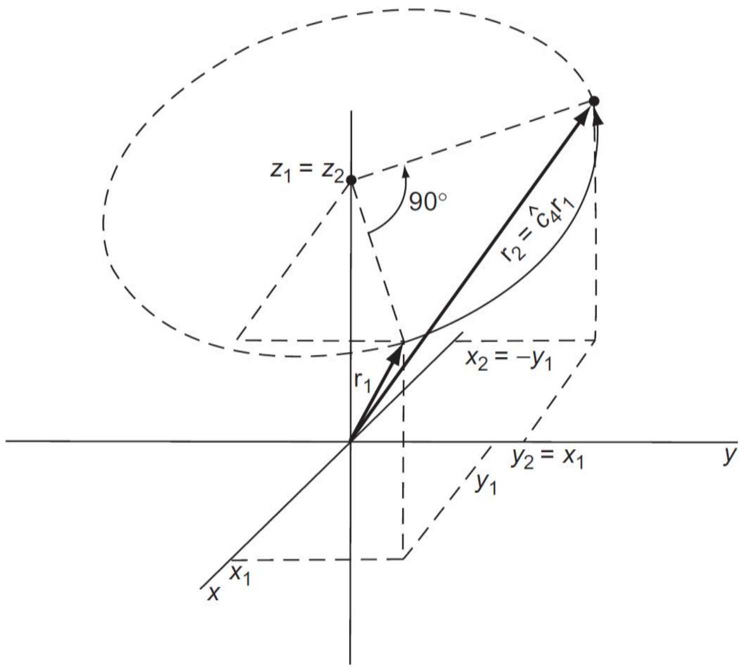
Proper Rotation Operators $\hat{\boldsymbol{C}}_{\boldsymbol{n}}$
(단순 회전 연산자)

Counterclockwise rotation by $2\pi/n$

- n applications of the rotation operator = 1 full rotation
- The rotation axis must be specified

$$\hat{C}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, z_1)$$

Action of $\widehat{C}_{4(z)}$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.2)

Not in Mortimer

Rotation about the z Axis

$$\hat{C}_{n(z)}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$x_2 = \left(\cos\frac{2\pi}{n}\right)x_1 - \left(\sin\frac{2\pi}{n}\right)y_1$$

$$y_2 = \left(\sin\frac{2\pi}{n}\right)x_1 + \left(\cos\frac{2\pi}{n}\right)y_1$$

$$z_2 = z_1$$

Improper Rotation Operators $\hat{\boldsymbol{S}}_{\boldsymbol{n}}$

(반사 회전 연산자)

$\hat{C}_n$ followed by $\hat{\sigma}$

(through a plane perpendicular to the rotation axis)

- The rotation axis must be specified

$$\hat{S}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, -z_1)$$

Examples 13.10 and 13.11

Find $\hat{C}_{4(z)}(1,-4,6)$ and $\hat{S}_{4(z)}(1,2,3)$.

Since $\hat{C}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, z_1)$,

$$\hat{C}_{4(z)}(1,-4,6) = (4,1,6)$$

Since $\hat{S}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, -z_1)$,

$$\hat{S}_{4(z)}(1,2,3) = (-2,1,-3)$$

Symmetry Elements (대칭 요소)

Point, line, or plane relative to which the symmetry operation is performed

- Must include the origin
- Sometimes denoted by the same symbol

- Inversion i: the origin
- Reflection σ: the reflection plane
- Rotation C_n, S_n: the rotation axis

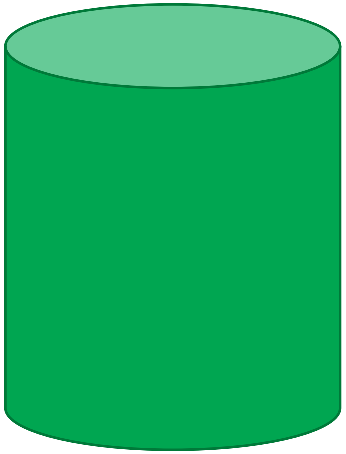
Symmetry of Object

- If after a symmetry operation, all particles in the object is in the same conformation as before, the symmetry operator belongs to the object.

- Coordinate system
  - origin: center of mass of the object
  - z axis: the rotation axis of highest order (largest n)

Example 13.12

List the symmetry operators belonging to a right circular cylinder.

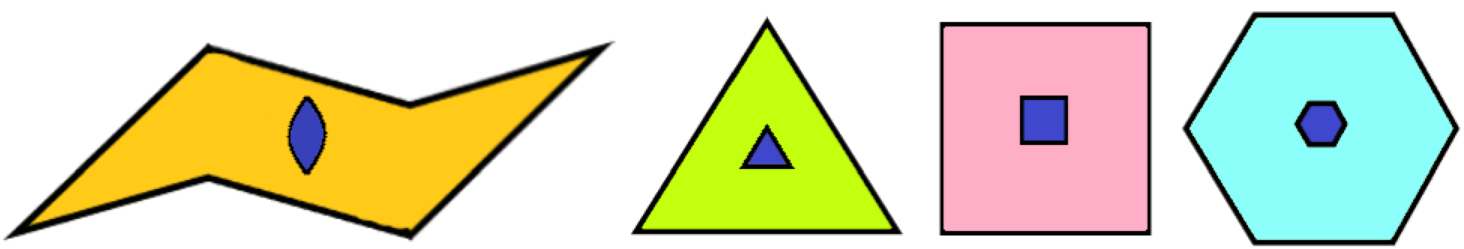


$\hat{\imath}, \hat{C}_{\infty(z)}, \hat{S}_{\infty(z)}, \hat{\sigma}_{\mathrm{h}}, \infty\hat{\sigma}_{\mathrm{v}}, \infty\hat{C}_2, \infty\hat{S}_2$

Not in Mortimer

Symbols for Rotational Symmetry

C_2 axis C_3 axis C_4 axis C_6 axis

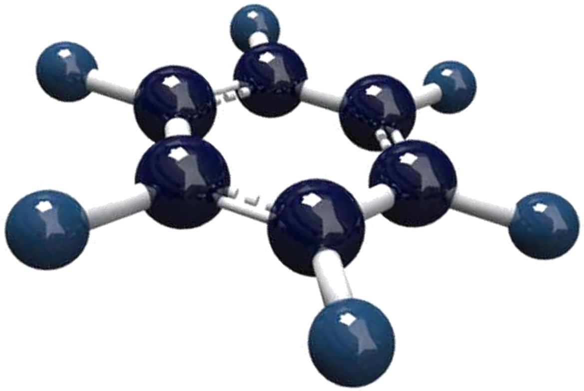


2-fold **3-fold** **4-fold** **6-fold**

(Image: https://bionerdnotes.wordpress.com/2020/06/01/symmetry-operations-and-its-nomenclature-for-the-space-groups-in-crystallography/)

Example 13.13

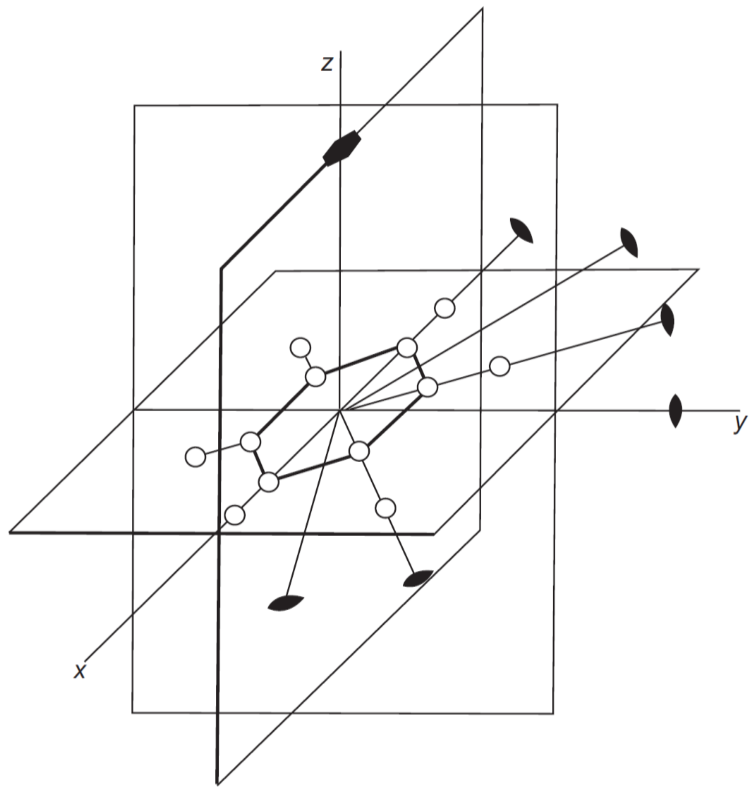
List the symmetry elements of the benzene molecule in its equilibrium conformation.



$\hat{\imath}, \hat{C}_{6(z)}, \hat{S}_{6(z)}, \hat{\sigma}_{\mathrm{h}}, 6\hat{\sigma}_{\mathrm{v}}, 6\hat{C}_2, 6\hat{S}_2$

Example 13.13

List the symmetry elements of the benzene molecule in its equilibrium conformation.



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.3)

Symmetry Operators on Functions

If an operator $\hat{O}$ operates as $\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$, for a function f,

$$\hat{O}f = g$$

where $g(x_2, y_2, z_2) = f(x_1, y_1, z_1)$.

Use the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2)!

Example 13.14

The normalized $2p_x$ wave function for the electron in a hydrogen atom is

$$\psi_{2px}(x,y,z) = x\exp\left[-\frac{(x^2+y^2+z^2)^{1/2}}{2a_0}\right],$$

where a_0 is the Bohr radius. Find $\hat{C}_{4(z)}\psi_{2px}$.

Since $\hat{C}_{4(z)}(x_1,y_1,z_1) = (x_2,y_2,z_2) = (-y_1,x_1,z_1)$, convert

$$x_2 \leftarrow -y_1,\ y_2 \leftarrow x_1,\ z_2 \leftarrow z_1;$$

$$\psi_{2px}(x_1,y_1,z_1) = x_1\exp\left[-\frac{(x_1^2+y_1^2+z_1^2)^{1/2}}{2a_0}\right]$$

$$\left(\hat{C}_{4(z)}\psi_{2px}\right)(x_2,y_2,z_2) = y_2\exp\left[-\frac{(y_2^2+(-x_2)^2+z_2^2)^{1/2}}{2a_0}\right]$$

Example 13.14

The normalized $2p_x$ wave function for the electron in a hydrogen atom is

$$\psi_{2px}(x,y,z) = x\exp\left[-\frac{(x^2+y^2+z^2)^{1/2}}{2a_0}\right],$$

where a_0 is the Bohr radius. Find $\hat{C}_{4(z)}\psi_{2px}$.

$$\left(\hat{C}_{4(z)}\psi_{2px}\right)(x_2,y_2,z_2) = y_2\exp\left[-\frac{(y_2^2+(-x_2)^2+z_2^2)^{1/2}}{2a_0}\right]$$

Hence, in typical coordinates,

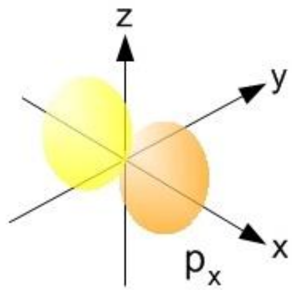
$$\hat{C}_{4(z)}\psi_{2px} = y\exp\left[-\frac{(x^2+y^2+z^2)^{1/2}}{2a_0}\right] = \psi_{2py}$$

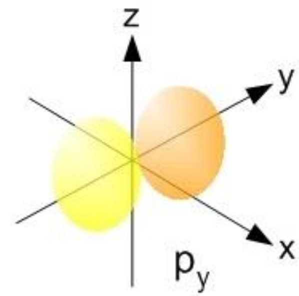
Example 13.14

The normalized $2p_x$ wave function for the electron in a hydrogen atom is

$$\psi_{2px}(x, y, z) = x \exp\left[-\frac{(x^2 + y^2 + z^2)^{1/2}}{2a_0}\right],$$

where a_0 is the Bohr radius. Find $\hat{C}_{4(z)}\psi_{2px}$.





(Image: https://commons.wikimedia.org/wiki/File:Single_electron_orbitals.jpg)

Eigenfunctions of Symmetry Operators

- Eigenvalue = +1

  the operator leaves a function unchanged

- Eigenvalue = -1

  the operator changes the sign of the function's value at all points

Exercise 13.14

(a) Find $\hat{\imath}\psi_{2px}$; (b) show that ψ_{2px} is an eigenfunction of the inversion operator; and (c) find its eigenvalue.

Since $\hat{\imath}(x_1, y_1, z_1) = (x_2, y_2, z_2) = (-x_1, -y_1, -z_1)$, convert

$$x_2 \leftarrow -x_1,\ y_2 \leftarrow -y_1,\ z_2 \leftarrow -z_1;$$

$$\psi_{2px}(x_1, y_1, z_1) = x_1 \exp\left[-\frac{(x_1^2 + y_1^2 + z_1^2)^{1/2}}{2a_0}\right]$$

$$\left(\hat{\imath}\psi_{2px}\right)(x_2, y_2, z_2) = -x_2 \exp\left[-\frac{\{(-x_2)^2 + (-y_2)^2 + (-z_2)^2\}^{1/2}}{2a_0}\right]$$

$$\hat{\imath}\psi_{2px} = -x \exp\left[-\frac{(x^2 + y^2 + z^2)^{1/2}}{2a_0}\right]$$

Exercise 13.14

(a) Find $\hat{\imath}\psi_{2px}$; (b) show that ψ_{2px} is an eigenfunction of the inversion operator; and (c) find its eigenvalue.

$$\hat{\imath}\psi_{2px} = -x \exp\left[-\frac{(x^2+y^2+z^2)^{1/2}}{2a_0}\right]$$

which reads that

$$\hat{\imath}\psi_{2px} = -\psi_{2px}$$

This is the eigenvalue equation, and it shows that ψ_{2px} is an eigenfunction of the inversion operator. The eigenvalue is -1.